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Assessing the nexus of Generative AI adoption, ethical considerations and organizational performance

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ABSTRACT

Numerous enterprises employ Generative AI (GenAI) for a plethora of business operations, which can enhance organizational effectiveness. The adoption might be driven by multiple factors influencing the business landscape. Additionally, numerous ethical considerations could impact the deployment of GenAI. This unique study investigated how organizations adopt GenAI and its effects on their performance. Further, this research utilized institutional theory and ethical guidelines for AI design to develop a research framework examining how organizations adopt GenAI and its impact on their performance. A survey of 384 managers from information technology (IT) and information technology-enabled services (ITeS) companies was conducted. Data analysis was done using PLS-SEM to examine and validate the proposed model. The study outcome reveals that institutional pressures, i.e., coercive, normative and mimetic forces, influence the use of GenAI in organizations. It was also found that fairness, accountability, transparency, accuracy and autonomy influence the use of GenAI. Also, the results divulge that the use of GenAI influences organizational performance and is moderated by organizational innovativeness. This study provides insights to developers of GenAI, senior management of companies, the government and IT policymakers by highlighting the institutional pressures and ethical principles influencing the use of GenAI.

1. Introduction

Generative artificial intelligence (hereafter, GenAI) is a distinctive technology that surged to prominence swiftly, captivating widespread attraction. GenAI is defined as - incorporating machine learning (ML) models to produce fresh content, such as text, images, audio, software code, video, and simulations, relies on extensive datasets employed for training the model (Budhwar et al., 2023). GenAI can generate new content in various forms, such as text and images (Susarla et al., 2023) using Natural Language Processing (NLP); however, it depends on the training of data timeliness, relevance and quality. This innovative GenAI is utilized and leveraged by companies for various business functions such as marketing, project management, analysis of data, customer relationship management, content creation, managing human resources, employee training and coding (Dwivedi et al., 2023; Gupta et al., 2023; Marr, 2023). Gartner, Inc. predicts that by 2026, over 80% of companies will have integrated GenAI through application programming interfaces or models or implemented GenAI-powered applications in production settings (Gartner, 2023).

Many companies use the ChatGPT (GenAI) launched by OpenAI. For example, *Slack*, a collaborative platform organization, uses ChatGPT to allow users to plan their workflow. Also, *Freshworks,* a software company, uses it to assist software engineers with coding (Marr, 2023). There are many GenAI tools creating hype in the market, such as Claude (Anthropic), ChatGPT (OpenAI), and Bard (Google) using large language models. Various industry sectors such as insurance, automobile, media, and IT are experimenting with GenAI. For example, IT and ITeS organizations are implementing GenAI, such as Git hub Copilot (Gupta et al., 2023).

GenAI has multiple benefits, such as providing on-time customer services, creating content for different business activities, managing human resources, providing instant customer feedback, providing

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product or service-related information, managing products, and digital marketing (Budhwar et al., 2023; Dwivedi et al., 2023; Gartner, 2023; Gupta et al., 2023; Ooi et al., 2023). Furthermore, GenAI can be used 24/7 as it is automated. It is evidenced that GenAI improved the efficiency of different business operations (Budhwar et al., 2023; Gupta et al., 2023; Marr, 2023; Raj et al., 2023; Tsigaris and Teixeira da Silva, 2023), GenAI minimizes the need for manual effort in routine tasks, freeing up employees to focus on strategic and complex responsibilities. It reduces the expenses of hiring additional staff for everyday operations and enables organizations to carry out various business processes with greater efficiency and effectiveness (Ooi et al., 2023). GenAI can ensure a competitive advantage by enhancing an organization's capability to manage customers and employees (Deloitte, 2023).

The present literature divulges why organizations need to use GenAI and mainly emphasizes the benefit of GenAI to organizations (Budhwar et al., 2023; Raj et al., 2023; Wamba et al., 2023). A study on the adoption of GenAI in the organizational setting has revealed various factors of the adoption of GenAI (Agrawal, 2023); however, there is a lack of research discussing the environmental and inter-organizational influences for the use of GenAI. The pressures from various business stakeholders compel organizations to adopt advanced technologies to improve business capabilities for organizational performance (Bag et al., 2021; Dubey et al., 2019b; Talaei-Khoei et al., 2024). There is a noticeable gap in research focused on the impact of GenAI on organizational performance, underscoring a call for scholars to pursue further exploratory work in this domain (Ooi et al., 2023; Agrawal, 2023b). Additionally, over the next decade, GenAI is forecasted to spur growth of $7 trillion in the global gross domestic product and contribute to an increase in productivity of up to 1.5% (Goldman Sachs, 2023). Considering the competitive edge offered by GenAI (Deloitte, 2023; Gupta et al., 2023), it is anticipated that environmental factors will influence the adoption of GenAI. Hence, it is crucial to investigate the environmental factors that affect the implementation of GenAI in organizations (Ooi et al., 2023).

Furthermore, though GenAI benefits individuals and organizations, some vital ethical concerns cannot be ignored (Akter et al., 2023; Díaz-Rodríguez et al., 2023; Heyder et al., 2023). As a pre-trained machine learning model employing unsupervised learning, GenAI processes vast datasets to generate responses that mimic human written language. Consequently, there is a risk that GenAI may be exploited, whether deliberately or inadvertently, for manipulative purposes (Chan, 2023). Utilitarianism poses ethical issues related to Gen AI, such as biases, plagiarism, and false text content (Tsou and Padgett Walsh, 2023). There have been incidences of bias, such as search biases and the use of biased data with flawed descriptions as input. Also, customer data privacy concerns companies, as it can be collected through data cookies. It could impact organizations' financial performance and public image (Cohen et al., 2023). As such, businesses need to be conscious of these matters and take steps to reduce the risks associated with using GenAI (Talaei-Khoei et al., 2024). The present studies discuss the ethical issues of using GenAI for business activities (Chan, 2023; Dwivedi et al., 2023; Matthew, 2023; Ooi et al., 2023). Additionally, it is anticipated that scholars will undertake further research into ethically responsible technological advancements (Stahl and Eke, 2024). The ethical considerations associated with sophisticated technologies, including issues of fairness, accountability, and transparency, are of paramount importance (Shin, 2021). Nevertheless, there appear to be no exploratory studies addressing the ethical aspects that influence the use of GenAI within organizations for business purposes (Stahl and Eke, 2024). Reflecting on these points, the following research question is posed:

RQ: *What environmental and ethical factors influence the use of GenAI in organizations and its contribution to organizational performance?*

The lens of the institutional theory (INST) (Dimaggio and Powell, 1983) and FAT (Fairness, accountability, and transparency) ethical principles of AI design (Kieslich et al., 2022; Shin and Park, 2019) is used to resolve these questions. The INST proposes the significant influence of environmental factors (Dubey et al., 2019b) and sheds light on the adoption factors of GenAI (Bag et al., 2021). The FAT (*Fairness, Accountability and Transparency*) ethical principles provide insights into ethical concerns related to the use of AI technology (Kieslich et al., 2022; Shin and Park, 2019). Also, two more ethical principles of AI design, *accuracy* and *autonomy* (Kieslich et al., 2022) are considered; hence, further called as FATAA principles of AI design. Additionally, whenever any advanced technology is used in the organization, organizational innovativeness (Marchiori et al., 2022) plays a crucial role in adoption. Hence, this research explores the effect of organizational innovativeness on organizational performance. This novel research addresses the existing gap and enhances the existing body of knowledge on the ethical use of GenAI and its impact on business performance. This research also primarily provides the influence of institutional forces on the use of GenAI. Additionally, this research offers crucial insights for marketers, managers, and developers regarding the ethical implementation of GenAI to strengthen organizational performance.

Further, this manuscript comprises a review of the literature, a section on theoretical foundations, and the formulation of hypotheses. Subsequently, the details of the operationalization of the constructs and the research methods used, along with the data analysis procedure and findings, are discussed. Next, the discussion section and theoretical and practical implications are presented. The paper ends with concluding remarks and suggestions for future research avenues.

2. Literature review

2.1. GenAI

An example of GenAI i.e., "Generative Pre-trained Transformer language (GPT) models (particularly GPT-3 and GPT-4) is a large-scale neural network-based language generation model trained on a wide range of Internet-based texts" (Stahl and Eke, 2024, p.2). GenAI called ChatGPT has become popular rapidly after its public release in November 2022 by the OpenAI. ChatGPT gained instant popularity due to its human-like ability to answer most questions, expel inappropriate questions, and accept mistakes. The language models in GenAI enable it to generate completely new content based on prompts provided by users (OpenAI Blog, 2022). This textual form of content can take different output forms, such as movie script, software code, poetry, social media content, marketing campaigns and research manuscripts. The quality output from GenAI depends on the user's input, as GenAI uses massive training data to provide this output (Deloitte, 2023; Gupta et al., 2023; Agrawal, 2023a).

GenAI has transformed the way businesses operate and engage with customers and employees. Existing research explores the application of GenAI across various business sectors, including manufacturing, entrepreneurship, information technology, tourism, and government (Wamba et al., 2023). ChatGPT 1 was launched in 2018, and ChatGPT 4 was launched in 2023 by the OpenAI. GenAI employs complex machine learning models to create original content, such as videos, audio, software code, and images. These models rely on extensive datasets for training. The timeliness, relevance, and quality of the training data are crucial in determining the contextual accuracy of the content produced by GenAI (McKinsey and Company Featured Insights, 2023). For example, GPT-4 utilizes a dataset comprising 45 terabytes of information gathered from the Internet up to 2021 for its training, which means its responses may not reflect the most recent and updated information. Additionally, using ChatGPT presents challenges such as the risk of data exposure, adherence to international laws and open-source licenses, privacy concerns, and the complexities of intellectual property rights (Cuomo, 2023).

Businesses are increasingly adopting GenAI; one notable example is Microsoft's investment in OpenAI, which aims to incorporate OpenAI's

capabilities into Microsoft Excel and Word. Microsoft's search engine and Bing also utilize GenAI to enhance search results. For code creation, GitHub Co-Pilot leverages Codex, a GenAI model, to assist with software development tasks (Ty Roush, 2023). Organizations reap multiple benefits from employing GenAI, including assistance with writing emails, providing personalized feedback, interacting with customers, managing time, aiding in coding tasks, enhancing customer service, and many other applications (O'Sullivan, 2023). The literature presents only a handful of studies exploring the adoption of GenAI (Brüns and Meißner, 2024; Budhathoki et al., 2024; Sidaoui et al., 2024).

2.2. Theoretical Background

2.2.1. Institutional theory (INST)

The INST (Dimaggio and Powell, 1983) is instrumental when an organization needs to adopt new technologies or practices (Bag et al., 2021). INST is grounded in strong social system dimensions, which include the policies and norms that provide the authoritative directives for social action (Scott, 2004). The INST discusses the integration of three forces i.e., mimetic, coercive and normative forces. INST suggests that environmental troops can significantly impact organizational actions (Dubey et al., 2019b). These forces arise from the government, culture, and industry, influencing organizations' operations and structure. Moreover, institutional pressures guide organizations to conform to the institutional environment, speeding up alignment and integration at the inter-organizational level (Dubey et al., 2019b; Tsai et al., 2013).

Coercive pressure is the pressure developed on organizations due to government and legally authorized rules and regulations to change practices or adopt technology (Bag et al., 2021; Dubey et al., 2019). The professional affiliations and codes in which top management and affiliate groups of customers and suppliers develop the pressure on the organization to change or update as per the norms are considered *normative pressures* (Messerschmidt and Hinz, 2013). Further, *mimetic pressure* discusses the actions of organizations that mimic the industry and competitors (Dubey et al., 2019). The INST acknowledges that technology adoption decisions encompass more than just logical considerations. It emphasizes that organizations are social entities influenced by a set of institutional logics that are internal as well as external in nature (Hartley et al., 2022; Messerschmidt and Hinz, 2013; Zheng et al., 2013). Furthermore, the INST has been used in technology adoption studies such as big data (Dubey et al., 2019a; Gupta et al., 2020), e-government (Zheng et al., 2013), digital supply chain (Gupta et al., 2020), RFID (Radio-frequency identification) (Tsai et al., 2013) and blockchain (Hartley et al., 2022; Shahzad et al., 2024). Hence, this work used INST to investigate the influence of environmental factors on the use of GenAI.

2.2.2. FATAA Ethical Principles

The ethical principles or codes provide the guidelines for the ethical requirements of AI systems (Guizzardi et al., 2020). The functional and quality needs of stakeholders regarding the AI system in line with ethical principles are considered as AI ethical requirements (Balasubramaniam et al., 2023). The ethical principles related to AI have been extensively argued by scholars (Kieslich et al., 2022; Shin, 2022; Shin and Park, 2019). Also, the research related to GenAI (ChatGPT) discusses ethical issues (Casal and Kessler, 2023; Lund et al., 2023; Rahimzadeh et al., 2023; Ray, 2023; Stahl and Eke, 2024).

GenAI systems, such as ChatGPT, have garnered significant interest for their vast advantages to people and organizations in easily executing various tasks. Yet, any new AI technology is evaluated for its advantages and ethical implications (Sidaoui et al., 2024). For example, the GenAI text results are challenging to differentiate from human output (Stahl and Eke, 2024) and can create organizational or employee communication issues. There are concerns regarding GenAI, such as eliminating the human element from essential decisions, privacy desecrations, and unfair or biased outcomes (Matthew, 2023).

Considering AI ethical principles when addressing the associated ethical issues is essential. These principles primarily consider the legal ramifications (Kieslich et al., 2022). The ethical principles/concerns such as accuracy, accountability, transparency, fairness, autonomy, security, and privacy are discussed about AI and GenAI technology (Díaz-Rodríguez et al., 2023; Ray, 2023; Song et al., 2022; Stahl and Eke, 2024). The ethical principles of FAT (*Fairness, accountability and transparency*) are crucial for using AI technology (Shin and Park, 2019). Hence, this research has considered these FAT ethical principles when examining the use of GenAI. Further, two critical ethical principles related to GenAI are used, i.e., *accuracy* (Kieslich et al., 2022) and *autonomy* (Kieslich et al., 2022). Hence, it is called FATAA ethical principles that are essential in the context of studying GenAI.

The *fairness* of GenAI output is crucial for its use, and if it is perceived as unfair, it is difficult for organizations to use it (Acikgoz et al., 2020). An AI system is regarded as fair when developing its algorithmic decision models is a collaborative effort with individuals from the organization rather than merely relying on the AI system's algorithms (Nagtegaal, 2021). The *accountability* of AI decisions or outputs is still questionable. The text results provided by GenAI may have limited validity due to pre-trained data or algorithms. The designers and developers develop AI systems and may not understand the ethical essence of the data output as the senior management directs them to create the AI systems (Shin, 2021). Hence, the accountability of AI systems' output is still not clear. If the AI system accountability is clear, organizations can use AI systems. The *transparency* of GenAI input data and, how outputs are provided to the user, the appropriateness of the outcome or decisions made by GenAI need to be transparent to the user (Balasubramaniam et al., 2023; Felzmann et al., 2019).

The *accuracy* of the AI system is precise and, as expected by the user, the user tends to use it (Song et al., 2022). The accuracy of AI systems is discussed as the ability to provide quick and correct output (Huang and Chueh, 2021). Therefore, AI's accuracy is essential for society's benefit (Kieslich et al., 2022). Further, AI's autonomy means AI's ability to conduct tasks without human intervention (Dwivedi et al., 2019; Hu et al., 2021a). *Autonomy* is one of the concerns that mentions that AI systems work without human supervision. Hence, identifying unethical results or issues remains the responsibility of the AI system itself. However, users accept AI system autonomy (Kieslich et al., 2022). This work explores the influence of FATAA on the use of GenAI.

2.3. Proposed conceptual framework and hypotheses formulation

The studies claim that using GenAI contributes to organizational performance (Chu, 2023; Wamba et al., 2023). However, organizations cannot achieve organizational performance if they fail to respond to environmental and external pressures (Aydiner et al., 2019). Also, environmental forces are pertinent when an organization adopts a new technology such as GenAI (Fosso Wamba et al., 2016; Wamba et al., 2023). Hence, we considered INST to study the influence of environmental forces such as normative, mimetic and coercive pressures (Hartley et al., 2022; Messerschmidt and Hinz, 2013) on the use of technology and, consequently, organizational performance.

Additionally, it has been emphasized that organizational innovativeness is a critical element in the adoption of novel technologies and in attaining improved organizational outcomes (Marchiori et al., 2022; Pesämaa et al., 2013). Organizational innovativeness is described as an organization's capacity to effectively utilize and integrate its resources and capabilities, transforming and reshaping them to generate novel and innovative outcomes (Marchiori et al., 2022). More innovative organizations are inclined to adopt cutting-edge technologies like GenAI, which can enhance organizational performance. Recent research has also drawn attention to the moderating influence of innovativeness (Acar and Ozs, 2018; Rubera et al., 2012; Song and Xie, 2000). Hence, we consider organizational innovativeness as a moderator between the use of GenAI and organizational innovativeness.

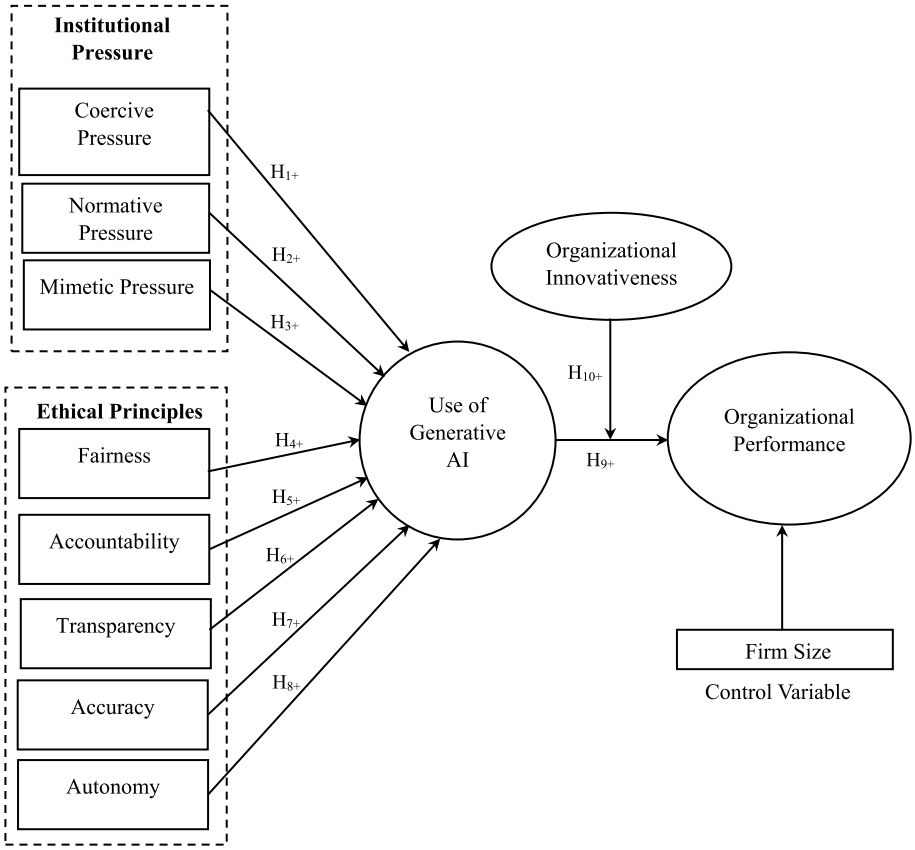


Fig. 1. Proposed research model (Dimaggio and Powell, 1983; Shin and Park, 2019).

Additionally, given that GenAI is a sophisticated technology powered by machine learning models, scholars have raised numerous ethical concerns that could affect its use (Díaz-Rodríguez et al., 2023; Ray, 2023; Song et al., 2022; Stahl and Eke, 2024). Hence, we explore the ethical principles such as FATAA influencing the use of GenAI (Kieslich et al., 2022; Shin and Park, 2019). Firm size is accounted for as a control variable, consistent with its use in prior information systems research (Hong and Zhu, 2006; Sivathanu, 2019a). The control variable is considered as a dummy variable. The organization size is derived from the number of employees in organizations: 1) Small and medium enterprises (SMEs) (with 2–999 employees), and 2) Large enterprises (with more than 999 employees). The proposed research model is portrayed in Fig. 1.

2.4. Hypotheses formulation

Coercive pressure emerges from the norms of professional networks and industry and government regulations (Ciftci et al., 2019; Fosso Wamba et al., 2016). Furthermore, the formal and informal pressures imparted on organizations by another organization are considered coercive pressure (Liang et al., 2007). The pressures developed from industry and government regulations influence the changes in business processes and the acceptance of new advanced technology (Dubey et al., 2019a). The existence of coercive pressure shaping the adoption of innovative technology is recognized, particularly concerning advanced technological solutions such as big data (Bag et al., 2021), blockchain (Hartley et al., 2022) and RFID (Fosso Wamba et al., 2016). The report from various IT companies' associates and global consulting firms imposes the use of GenAI in organizations to improve the performance of organizations (Deloitte, 2023; Gupta et al., 2023; Mckinsey & Company, 2023; NASSCOM, 2023). Prominent consulting firms such as McKinsey, Gartner, and Deloitte advocate for the use of ChatGPT and its advantages; therefore, coercive pressure may drive the adoption of GenAI within organizations. Hence, it is hypothesized that:

H1. Coercive pressure positively influences the use of GenAI by organizations.

Normative pressure originates from customer expectations and professional standards/codes, necessitating compliance with detailed guidelines (Dubey et al., 2019a; Sarkis et al., 2011). Research substantiates the effect of normative pressure in transforming business procedures and fostering the acceptance of novel technologies (Bag et al., 2021, 2023; Hartley et al., 2022). Clients and customers of IT firms have begun to utilize GenAI and anticipate that these firms will integrate GenAI into diverse business operations and communication practices (Baruah, 2023; Deloitte, 2019). IT company employees use GenAI to write codes and demand the use of GenAI to improve their productivity (Cohen et al., 2023; Sharma, 2023). The pressures from clients, employees, and professional institutions are believed to motivate organizations to implement GenAI. Hence, we propose the following hypothesis:

H2. Normative pressure positively influences the use of GenAI by organizations.

When organizations are unclear about the highly dynamic environmental changes and technological advancements, they tend to copy their competitors (Bag et al., 2023). Organizations make logical decisions in response to technological uncertainty by weighing the advantages and practical applications of the technology. The propensity of an organization to replicate the behaviour of its competitors or other businesses is identified as mimetic pressure. It happens when an organization does not understand the technology (Dubey et al., 2019). GenAI is a new innovative technology that has emerged, and organizations are feeling pressure from competitors to use Gen AI (Dwivedi et al., 2023). GenAI's

rapid adoption by numerous users is due to its capacity to deliver immediate outcomes in areas like programming, customer service, and content generation, which in turn draws many organizations to utilize it (Deloitte, 2019, 2023). Hence, organizations might adopt GenAI to mimic the competitors; therefore, the following hypothesis is proposed:

H3. Mimetic pressure positively influences the use of GenAI by organizations.

Fairness of AI technology means every individual deserves equitable treatment, and the deployment of AI should not lead to prejudicial discrimination against any person, community, or collective (Ashok et al., 2022). It is widely discussed that GenAI algorithms and development should not be biased, though the machine learning models may have biased social data (Martin, 2019). It is argued that AI technology developers should ensure fairness and social justice (Díaz-Rodríguez et al., 2023). On the contrary, it is also discussed that AI's emancipatory power can help develop a failed and unbiased society (Ashok et al., 2022; Martin, 2019). Research has shown that GPT-3 can produce biased results, which may inadvertently or deliberately lead to discrimination against certain social groups (Chan, 2023). If GenAI ensures outcomes that are fair, unbiased, and devoid of prejudice, it would likely be integrated into organizational use (Shin, 2021). Thus, the following hypothesis is proposed:

H4. Fairness positively influences the use of GenAI by organizations.

Accountability is defined as the ethical consideration of accountability regarding the functionality of AI (Floridi et al., 2020). There is a significant discussion concerning who is accountable for decisions that arise from algorithmic processes, mechanical systems, and the choices made by the software engineers who create them (Ashok et al., 2022). The accountability for outcomes produced by GenAI is debatable since it delivers results based on its programming and the pre-existing data from a specific time frame. Also, the code crafted by developers can sometimes yield erroneous results due to either deliberate or inadvertent outcomes (Elliott et al., 2021). This research considers accountability as the ethical responsibility of GenAI providers to ensure the production of ethical outcomes (Chan, 2023). This responsibility goes to various stakeholders involved in developing and using GenAI. The assurance of accountability for the outputs given by GenAI may increase user confidence in utilizing GenAI technologies. Therefore, the following hypothesis is posited:

H5. Accountability positively influences the use of GenAI by organizations.

The transparency of AI encompasses the specifics of service logic, various forms of data handling, including sensitive information, and potential implications arising from the system's implicit acquisition of user knowledge (Shin and Park, 2019). Transparency is also seen as enabling users to understand the system's operations and rationale, to recognize the decisions or outcomes it generates, and to appraise the adequacy of these decisions or results (Felzmann et al., 2019). Moreover, it is argued that the "black box" nature of AI algorithms relates to their complexity and the challenge presented for individuals in comprehending the internal workings of these algorithms (Elliott et al., 2021; Shin and Park, 2019). In the context of GenAI, which is pre-trained on diverse datasets from the Internet using intricate machine learning models, organizations need to understand the data gathering and processing methodologies to deliver outcomes or make decisions (Matthew, 2023). Scholars argue that transparency can impact the use behaviour of AI technology (Balasubramaniam et al., 2023; Felzmann et al., 2019). The transparency regarding the input data and the processing methods used by GenAI to provide outputs, as well as the appropriateness of the data, should be made clear for organizations considering the use of GenAI for various business functions. Consequently, the following hypothesis is formulated:

H6. Transparency positively influences the use of GenAI by organizations.

Accuracy refers to the capability of GenAI to deliver optimized outcomes or forecast results that are informed by algorithms and based on pre-trained data (Shin, 2021). Accuracy is also viewed as the timeliness and precision of output/communication that aligns with the user's anticipations (Song et al., 2022). Since the outputs generated by GenAI rely on pre-trained data and algorithms, there is a possibility that inaccuracies in the algorithms or the pre-training datasets could lead to results from GenAI that may not be precise. Accuracy is an important parameter that users consider to accept AI technology (Gursoy et al., 2019). However, GenAI technology cannot validate data accuracy (Dwivedi et al., 2023).

Additionally, GenAI can generate results from minimal text inputs, as it excels in identifying keywords and producing relevant outcomes (Dwivedi et al., 2023). Users could potentially perceive the instant outcomes from GenAI as accurate, mistaking response speed for accuracy. Hence, the following hypothesis is proposed:

H7. Accuracy positively influences the use of GenAI by organizations.

GenAI is equipped with features of autonomy, which allow GenAI to autonomously perform tasks by sensing, thinking, and acting upon inputs and provide sensible outputs that the user expects (Dwivedi et al., 2019; Hu et al., 2021a). The autonomy of GenAI is characterized by its capacity to generate significant responses and independently arrive at decisions (Chan, 2023). GenAI's ability to engage with users efficiently and swiftly accomplish tasks aligned with users' expectations enhances its autonomous features and encourages user adoption (Beer et al., 2014). Nevertheless, GenAI is susceptible to exploitation by malicious actors who could misuse it to create and disseminate misinformation, which would be considered ethically wrong (Chan, 2023). The autonomous qualities of GenAI can create a user experience that feels intuitive and relatable, potentially increasing user comfort and driving adoption (De Smet et al., 2023). Hence, the following hypothesis is proposed:

H8. Autonomy positively influences the use of GenAI by organizations.

Technology is improving business performance and several studies argue that the use of technology helps in improving organizational performance (Marchiori et al., 2022; Mariani et al., 2023; Parteka and Kordalska, 2023; Pillai and Srivastava, 2023). The organization's performance is discussed as the degree to which the organization provides excellent services to customers, manages resources, utilizes resources efficiently and reduces customer complaints (Marchiori et al., 2022; Pesämaa et al., 2013). GenAI significantly aids organizations in addressing inquiries and offering feedback to employees and customers. In IT firms, staff can enhance their coding abilities by utilizing GenAI. Moreover, GenAI can quickly furnish stakeholders with information to bolsters business operations' effectiveness (Chu, 2023; Wamba et al., 2023; Raj et al., 2023). The ChatGPT research also confirms its contribution to organizational performance (Chu, 2023; De Smet et al., 2023). Consequently, the following hypothesis is proposed:

H9. Use of GenAI in organizations influences organizational performance.

In the current demanding business context, it is imperative for organizations to engage in innovation to elevate their organizational performance (Marchiori et al., 2022; Pesämaa et al., 2013). Organizational innovativeness refers to the capacity of an organization to create and support the development of novel processes, services, and products (Calantone et al., 2002; Marchiori et al., 2022). Research suggests that applying technology across various business activities contributes to enhancing organisational performance (Marchiori et al., 2022; Pillai and Srivastava, 2023; Agrawal, 2023b). The strategic importance of advanced technology implies that organizations inclined towards innovation are likely to experiment with new technologies across business

operations to attain organizational performance. Consequently, innovativeness acts as a moderating factor in the relationship between the deployment of technology and organizational performance (Acar and Ozs, 2018; Song and Xie, 2000). IT and ITeS companies should explore new and emerging advanced technologies like GenAI. Being innovative prompts organizations to adopt such novel technologies in pursuit of enhanced organizational performance. Thus, the following hypothesis is proposed:

H10. Organizational innovativeness moderates the relationship between the use of GenAI and organizational performance.

3. Research methodology

3.1. Survey measures

To validate the theoretical model and test the hypotheses, a comprehensive survey was conducted. The questionnaire included items on constructs considered for this study, demographic questions relating to work experience, organization size, education, and prior experience with AI or similar technology. Coercive, normative and mimetic pressures were evaluated using a three-item scale for each validated by prior research (Dubey et al., 2016; Gupta et al., 2020). Fairness was gauged using a three-item scale adapted from Shin (2021) and Shin and Park (2019). A three-item scale was used to measure accountability and transparency, validated by Shin (2021) and Shin and Park (2019). For measuring accuracy, a three-item scale was adapted from Song et al. (2022). A four-item scale validated by Hu et al. (2021b) was used to measure autonomy. A three-item scale validated by Lin et al. (2018), Pillai et al. (2021), and Sackey and Bester (2016) was used to measure the use of GenAI. Innovation Pesämaa et al. (2013) and organizational performance were gauged using a four-item scale adapted from (Calantone et al., 2002; Marchiori et al., 2022).

To avoid unambiguousness and improve the clarity of items used in our study, pretesting was done. As (Willis, 2016) suggested pre-testing should be done even for the standardized questionnaire as this helps eliminate any possible ambiguity and makes the respondents interpret the items as intended. A sample of 20 respondents participated in the pre-testing to confirm the intent and clarity of the survey. Developing on the potential understanding obtained from the pre-testing, a pilot study was conducted successively with a sample of 50. This number is considered significant for the pilot study. After that the data collection for the significant pool of respondents started after receiving encouraging results from the pilot study.

To prevent potential biases that could arise because of the data collection technique or the design used for research, a three-wave time-lagged approach was used. All the participants were again assured about their responses' anonymity and confidentiality. A unique code was given to all the participants, which helped in data tracking across three waves.

Data was collected from senior managers, project heads, and CTOs working in IT and IteS companies in India. Out of the 528 initially approached participants, 489 consented. During the first wave (T1), information on trust, autonomy, fairness, accuracy, accountability, coercive pressure, normative pressure, and mimetic pressure were collected. From the 489 questionnaires distributed, 462 responses were gathered. After a two-week interval, in the second wave (T2), questions on the use of GenAI and innovativeness were presented to the initial 462 respondents. It resulted in 417 completed responses. After another two-week break, the third wave (T3) was initiated, targeting the 417 respondents from the previous phase, focusing on demographic details (i.e., experience, size of the organization, education, and previous experience with AI or similar technology) and organizational performance. At the end of these three waves, the cumulative total amounted to 384 responses. Non-response bias is not a concern, as shown by the result of the *t*-test ($p = 0.41$) (Tsou and Hsu, 2015).

3.2. Common Method Bias (CMB) and endogeneity

The single factor Harman's test was conducted and 24.71% was variance (Wang et al., 2018). Hence, CMB is not a concern. Furthermore, CMB is not observed as values of multicollinearity VIF (Liang et al., 2007) are below '5' (Kock, 2017) as mentioned in Table 1. The structural model recursively may lead to endogeneity (Dubey et al., 2018). To evaluate the endogeneity of the structural model and to confirm the reliability and validity of the model, Ramsey regression equation error test (Lai et al., 2018) is done. Another suggested method to reduce the impact of method bias involves employing a marker variable, such as Fashion Consciousness (FCS) (Simmering et al., 2015), which has no conceptual relation to the constructs in the proposed model (Chin et al., 2012; Rönkkö and Ylitalo, 2011; Shiau and Chau, 2016). The examination of the marker variable (i.e.FCS) likewise indicated the absence of common method bias.

4. Data analysis

Our study employed PLS-SEM as it is considered and known for its ability to vigorously test models conducive to theory building (Hair et al., 2011). Further, PLS-SEM is suitable for studies that are characterized by a forecasting mindset, intricate model structures, and the use of formatively assessed constructs (Ali et al., 2018). Moreover, as our study included the analysis of 36 items spanning over 11 constructs and its expansive sample, PLS-SEM was deemed appropriate for this analysis. PLS-SEM has been used in various AI technology acceptance research (Chen et al., 2023; Sohn et al., 2021). The Kurtosis and Skewness values remained between '-2' and '+2', thereby reinforcing its normality. Also, the maximum variance explained by the data was found to be less than 50% (Podsakoff et al., 2003).

4.1. Measurement model

Convergent validity, reliability, and discriminant validity were tested to access the measurement model for all the constructs present in the study. The average variance extracted (AVE) value was checked for all the constructs for convergent validity. All the values were found to be above the threshold of 0.50. To check the internal consistency, Cronbach's alpha (α) and composite reliability were also computed (for all constructs). The result (see Table 1) indicates that all the values were within the specified limit, which signifies the quality criteria of 'first-order' constructs involved in this study. Cronbach's alpha (α) values for all the constructs were above the threshold of 0.70. Heterotrait–Monotrait (HTMT) ratio of correlation was investigated to test the discriminant validity. The HTMT ratios of all the constructs were less than the threshold value of 0.85, revealing that the discriminant validity of all the constructs is appropriate (see Table 2).

4.2. Structural model

To measure the predictive accuracy of the model used in this study, the coefficient of determination (R^2 coefficient) was explored. The values of R^2 coefficient for all the variables involved in this study were significant. The R^2 coefficient values for the use of GenAI were 0.740 and organizational performance as 0.370. The adjusted R^2 coefficient values for the use of GenAI and organizational performance were found as 0.735 and 0.365, respectively. The variance inflation factor (VIF) of all the variables involved in this study was tested to check the collinearity issue. The VIF values for all the items were within the cut-off limits (Ringle et al., 2015). The blindfold method was adopted to assess the Stone–Geisser's Q^2; this is assessed to confirm the model's predictive accuracy (Chin et al., 2020). As recommended, these values must be more than zero. The values of Q^2 for the use of GenAI and organizational performance are 0.720 and 0.408, respectively.

As mentioned in Table 3 and Fig. 2, *coercive pressure* was found to

Table 1
Operationalization of constructs.

Construct	Items	Factor loading	Source(s)
Coercive Pressure	The government demands that our organization should use GenAI.	0.837	(Dubey et al., 2016; Gupta et al., 2020)
AVE = 0.715	The industry association demands that our organization should use GenAI.	0.845	
CR = 0.892	The client associations demand that our organization use GenAI.	0.856	
α = 0.839			
VIF = 1.319			
Normative Pressure	Customers appreciate the use of GenAI.	0.876	(Dubey et al., 2016; Gupta et al., 2020)
AVE = 0.785	Clients of our organizations have adopted GenAI.	0.845	
CR = 0.893	The professional body's promotion of GenAI has influenced our organization to use GenAI.	0.874	
α = 0.801			
VIF = 1.158			
Mimetic Pressure	Our key competitors who use GenAI have benefitted extensively.	0.877	(Dubey et al., 2016; Gupta et al., 2020)
AVE = 0.746	Our key competitors who use GenAI are favourably perceived by other players within the same industry.	0.881	
CR = 0.883			
α = 0.826	Our key competitors who have used GenAI are favourably perceived by the customers.	0.864	
VIF = 1.249			
Fairness	GenAI does not perform any favouritism and discrimination against any individuals in our organization.	0.846	(Shin, 2021; Shin and Park, 2019)
AVE = 0.720			
CR = 0.861	GenAI outputs are benchmarked and identical for all individuals for specific matters.	0.875	
α = 0.844	GenAI follows the defined process impartiality with no bias.	0.849	
VIF = 1.264			
Accountability	Someone should monitor GenAI and be accountable for the adverse societal and individual effects.	0.855	(Shin, 2021; Shin and Park, 2019)
AVE = 0.748	GenAI algorithms should be designed to enable third parties to scrutinize and assess the response of an algorithm.	0.868	
CR = 0.878			
α = 0.803	AI algorithms should be able to alter a GenAI functioning in its complete configuration using only certain manipulations.	0.873	
VIF = 1.601			
Transparency	The criteria of evaluation of GenAI algorithms should be revealed to the public and ensure that people should understand it (Understandability).	0.864	Shin (2021)
AVE = 0.734			
CR = 0.835	The results/decisions made by GenAI should be explainable to the people who are particularly affected by those results and decisions. (Explainability)	0.856	
α = 0.807			
VIF = 1.567	The Gen AI algorithm function should let people know how well the internal states of ML models and algorithms can be understood from knowledge of its external outputs (Observability)	0.857	
Accuracy	GenAI instantly replies to all individuals in our organization.	0.863	Song et al. (2022)
AVE = 0.732	GenAI interacts accurately with all individuals in our organization.	0.845	
CR = 0.876	GenAI completely replies to all individuals in our organization.	0.859	
α = 0.816			
VIF = 1.418			
Autonomy	GenAI can autonomously provide choices to us for further courses of action.	0.872	Hu et al. (2021a)
AVE = 0.730	Independently, GenAI can offer recommendations for action plans for the cited problems and matters.	0.832	
CR = 0.846	GenAI can autonomously suggest the steps in the execution of the plan for the assigned matters/issues.	0.851	
α = 0.818	GenAI can autonomously recommend what can be done.	0.863	
VIF = 1.275			
Use of GenAI	Our organization proposes to use GenAI in the near future.	0.851	(Lin et al., 2018; Pillai et al., 2021; Sackey and Bester, 2016)
AVE = 0.736	Our organization is inclined to increase the use of GenAI.	0.849	
CR = 0.867	Our organization has the capacity to use GenAI.	0.873	
α = 0.823			
VIF = 1.682			
Innovativeness	Generally, managers in our organization emphasize innovativeness rather than providing established software and services.	0.821	Pesämaa et al. (2013)
AVE = 0.705			
CR = 0.856	Our organization has introduced many new services and software over the past few years.	0.833	
α = 0.823	Societal transformations often serve as a catalyst for our thinking, leading us to generate innovative concepts.	0.856	
VIF = 1.305			
	Our organization can easily come across ideas that have the potential to be transformed into services or developed as software.	0.849	
Organizational performance	Customers get excellent software and service from our organization.	0.871	(Calantone et al., 2002; Marchiori et al., 2022)
	Our organization efficiently manages public resources, ensuring their utilization.	0.865	
AVE = 0.752	There isn't really a lot that the clients or employees of our organization can find fault with.	0.855	
CR = 0.874	Overall, our organization is performing well.	0.878	
α = 0.832			

have a significant and positive impact on the use of GenAI (β = 0.149, t = 2.128, p = 0.036), supporting H_1. *normative pressure* was found to have a positive impact on the use of GenAI (β = 0.208, t = 2.602, p = 0.004), thereby supporting H_2. *mimetic pressure* was found to have a significant impact on the use of GenAI (β = 0.321, t = 4.012, p = 0.001), thereby supporting H_3. Further, *fairness* was found to have a positive impact on the use of GenAI (β = 0.288, t = 3.620, p = 0.002); hence, H4 is supported. *Accountability* positively influences the use of GenAI (β = 0.298, t = 3.725, p = 0.002), thereby supporting H5. *transparency* was also found to have a positive impact on the use of GenAI (β = 0.217, t = 2.712, p = 0.003), thereby supporting H6. Further, both accuracy and autonomy were found to exert a significant influence on the use of GenAI; hence, supporting H7 AI (β = 0.302, t = 3.823, p = 0.003) and H8 AI (β = 0.276 t = 3.067, p = 0.003). The use of GenAI was found to have a significant and positive impact on organizational performance (β = 0.348, t = 4.976, p = 0.000), thereby supporting H9. Innovativeness positively moderates the relationship between the use of GenAI and organizational performance (β = 0.167, t = 2.385, p = 0.042), thereby supporting H10. It is revealed the control variable – firm size does not have any influence on organizational performance (β = 0.037, NS)

(Sivathanu, 2018, 2019), contradicting the extant research (Hong and Zhu, 2006).

5. Discussion

This investigation offers an insightful framework for the use of GenAI and its influence on organizational performance using INST and FATAA ethical principles of AI design. Further, it highlights the moderation effect of organizational innovativeness on the association between the use of GenAI and organizational performance. The finding of this research reveals the influence of coercive, normative, and memetic pressure on the use of GenAI, confirming H1, H2 and H3, supports the extant advanced technology adoption studies (Dubey et al., 2019; Tsai et al., 2013). GenAI technology is useful for IT and ITeS companies to perform different business processes. The industry associations such as NAASCOM (NASSCOM, 2023) emphasize the use of GenAI. Many customers and clients of these organizations demand the use of GenAI. Further, many IT professional bodies and various business stakeholders, such as customers and clients, are using GenAI and expect IT and ITeS organizations to use GenAI. IT companies face tremendous pressure from competitors to use advanced technology (Deloitte, 2023; Gupta et al., 2023; Ty Roush, 2023). Hence, organizations face *coercive, normative*, and *memetic* pressure to use GenAI.

Further, the ethical principles of AI design, such as fairness, accountability, transparency, accuracy, and autonomy, influence the use of GenAI, confirming hypotheses H4, H5, H6, H7 and H8. Further, the results show the impact of fairness (H4) on the use of GenAI, confirming the extant result on AI ethics (Shin and Park, 2019). GenAI output or decisions should be unbiased and non-discriminatory and avoid favouritism. This fairness of GenAI outputs and results would lead to the use of GenAI. The findings indicate that accountability (H5) is imperative for its use. The providers of GenAI and the executives overseeing its implementation in organizations should be accountable for the results and decisions linked to GenAI usage. GenAI should be monitored by companies such as OpenAI constantly for results and decisions made, and periodic monitoring of the manipulation of GenAI should occur. If the GenAI developer companies take accountability, it would lead to better use of the GenAI in organizations.

Further, transparency (H6) of GenAI influences the use of GenAI (Balasubramaniam et al., 2023). If organizations have a clear understanding of GenAI operations, including its mechanics, the reasoning behind its actions, the development process of algorithms and language models (Felzmann et al., 2019) that produce the results, such openness can foster the confident utilization of GenAI. Moreover, the accuracy (H7) and autonomy (H8) of GenAI are crucial for its deployment in organizations. The results of GenAI should be as per the expectations of the organization and its stakeholders. Also, GenAI should be able to reply and interact with individuals in the organization instantly and accurately. GenAI performs the task autonomously and provides choices for future courses of action, generating a recommendation for the problems stated to GenAI and also autonomously recommending the solutions to the problems (Hu et al., 2021). This would attract organizations to use the GenAI.

Table 3
Structural model assessments.

H#	Path	β	*t*-statistics	Sig.
H_1	CPR → UGA	0.149	2.129**	0.036
H_2	NPR → UGA	0.208	2.602***	0.004
H_3	MPR → UGA	0.321	4.013***	0.001
H_4	FNS→ UGA	0.288	3.620***	0.002
H_5	ACN → UGA	0.298	3.725***	0.002
H_6	TRN → UGA	0.217	2.713***	0.003
H_7	ACR→ UGA	0.302	3.823***	0.003
H_8	ATN→ UGA	0.276	3.067***	0.003
H_9	UGA→ OPR	0.348	4.971***	0.001
H_{10}	UGA→ INV→ OPR	0.167	2.386**	0.042

[Note: t-values for two-tailed test: ***t-value 2.58 (sig. level:1%), **t-value 1.96 (sig. level: 5%), *t-value 1.65 (sig. level: 10%)]. CPR = Coercive Pressure, NPR = Normative Pressure, MPR = Mimetic Pressure, FNS = Fairness, ACN = Accountability, TRN = Transparency, ACR = Accuracy, ATN = Autonomy, INV = Innovation, UGA = Use of GenAI, INV = Innovativeness, OPR = Organizational performance.

Additionally, the use of GenAI enhances organizational performance (H9), corroborating the findings from previous research (Chu, 2023; Wamba et al., 2023; Raj et al., 2023). The use of GenAI in organizations helps to improve employee efficiency, provide better customer services and reduce customer complaints (Gupta et al., 2023), can assist in product development (Dwivedi et al., 2023; Ooi et al., 2023) and contribute to overall performance. The results depict that organizational innovativeness (H10) moderates the association between the use of GenAI and organizational performance (Acar and Ozs, 2018). GenAI can be used better in organizations if they have an innovative mindset and capabilities. Organizations' innovativeness directly emphasizes using innovative software, developing new software, and considering new ideas for transforming new software services (Pesämaa et al., 2013). Hence, organizations that are more innovative will experiment with advanced technology, software, and services such as GenAI.

Further, the control variable firm size is not found to be significant. Since the control variable firm size in this study is solely based on the number of employees within a firm, larger firms having more employees may use GenAI for organization performance; however, smaller organizations, despite having fewer employees, can still benefit from the use of GenAI by focusing on specific areas or task leading to notable organizational performance. Hence, in this study, the control variable firm size does not significantly influence organizational performance.

5.1. Theoretical implications

This research provides a distinctive theoretical model using INST and ethical principles of AI and investigates the factors influencing the use of GenAI and its contribution to organizational performance. Firstly, this study addresses the researchers' call to investigate the external factors

Table 2
Discriminant validity (HTMT).

	CPR	NPR	MPR	ACN	TRN	ACR	INV	UGA	OPR
CPR									
NPR	0.797								
MPR	0.706	0.813							
ACN	0.756	0.767	0.709						
TRN	0.435	0.323	0.368	0.292					
ACR	0.561	0.508	0.538	0.472	0.528				
INV	0.452	0.421	0.433	0.322	0.420	0.699			
UGA	0.716	0.805	0.773	0.702	0.370	0.502	0.375		
OPR	0.466	0.429	0.426	0.404	0.446	0.720	0.577	0.446	

[Note: ACN = Accountability, ACR = Accuracy, CPR = Coercive Pressure, INV = Innovativeness, MPR = Mimetic Pressure, NPR = Normative Pressure, OPR = Organizational performance, TRN = Transparency, UGA = Use of GenAI].

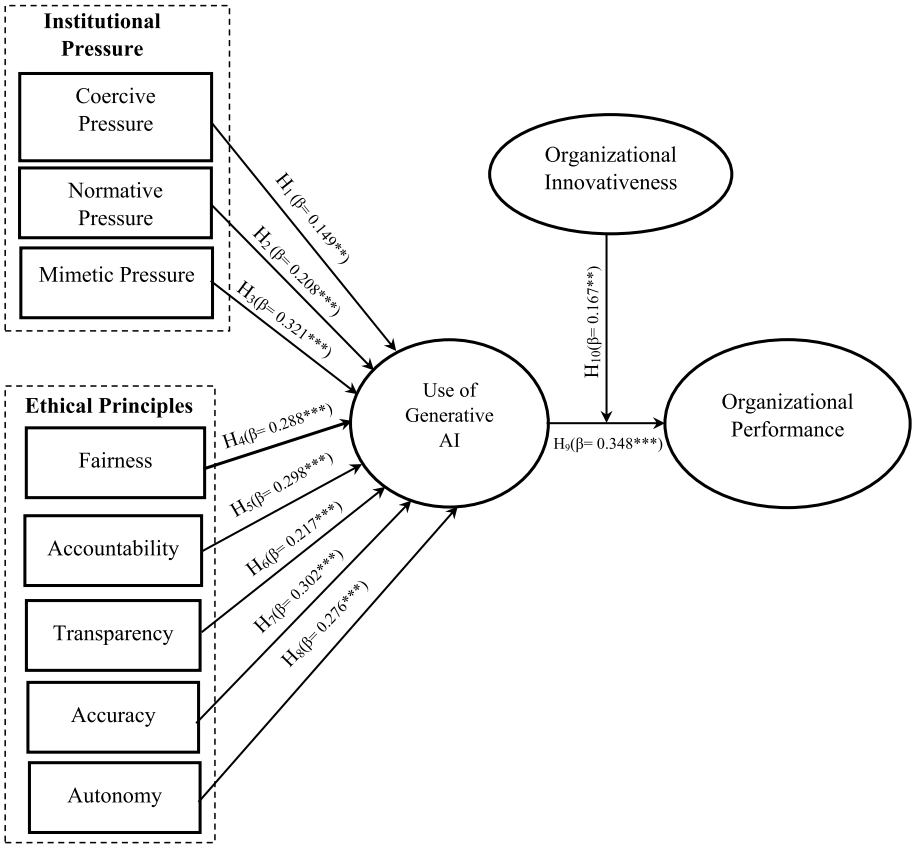


Fig. 2. Structural model.

influencing the adoption of innovative technologies, such as GenAI, using the framework of INST (Aydiner et al., 2019; Dwivedi et al., 2023). Since institutional pressures are crucial in the acceptance of new technology, this is the first research that presents concrete evidence of these pressures on the use of GenAI. Mimetic pressure is strong concerning the use of GenAI in IT and ITeS Companies compared to coercive and normative pressures.

The existing studies of technology acceptance have mainly used technology adoption models (Ameye et al., 2023); however, this research highlights ethical factors influencing the use of GenAI from an organizational perspective, which was not explored much in earlier AI technology adoption studies. Research into the application of GenAI and its ethical implications is still in the nascent stage (Wamba et al., 2023) and in high demand (Dwivedi et al., 2023; Stahl and Eke, 2024; Wong et al., 2023). The existing studies are qualitative and mostly discuss the experimentation of GenAI, its benefits, challenges and ethical concerns (Chan, 2023; Sidaoui et al., 2024). However, there is a lack of research as far as exploratory studies on the use of GenAI and ethical aspects of GenAI are concerned (Dwivedi et al., 2023; Stahl and Eke, 2024; Talaei-Khoei et al., 2024b). This research gap is answered by the empirical investigation using the FATAA factors influencing GenAI adoption at an organizational level and highlighting the functional features of GenAI for ethical use. This work agrees with the recent findings of prior studies (Shin, 2021; Shin and Park, 2019) and confirms the influence of transparency, accountability and fairness on the use of GenAI and, consequently, its contribution to organizational performance. Further, this work fills the research gap (Hu et al., 2021a; Shin, 2021; Shin et al., 2020) by exploring the role of autonomy and accuracy in the use of GenAI.

Notably, this study confirms that the use of GenAI leverages organizational performance (Talaei-Khoei et al., 2024b), which contributes to strategic business management and organizational performance literature. This work contributes to the theory by confirming the moderating role of organizational innovativeness on the relationship between the use of GenAI and organizational performance (Acar and Ozs, 2018). Organizations with greater innovativeness may possess an enhanced capacity and eagerness to incorporate and utilize GenAI in diverse business operations, which could consequently result in improved organizational performance. Overall, this study adds to the body of knowledge regarding cutting-edge technology by examining aspects like the adoption and ethical utilization of GenAI, along with its relation to organizational innovation and performance.

5.2. *Managerial implications*

This study offers strategic insights for managers of IT and ITeS organizations and senior executives across various companies. The research demonstrates how institutional pressures affect the use of GenAI and their substantial impact on organizational performance. It provides managers with insights into how coercive, normative, and mimetic pressures influence the use of GenAI in their practices. Managers can interpret this as the need to employ GenAI across different organizational functions to enhance their efficiency. Further, the mimetic pressure has a relatively higher magnitude of influence on the use of GenAI than other pressures; hence, organizations need to understand that to survive in this competitive environment, they must adopt GenAI. Managers should align GenAI use with industry best practices and standards to navigate the normative pressures. Al managers need to stay informed about legal/government requirements and use GenAI accordingly to stay compliant.

Further, this research highlights the moral concerns by revealing the impact of AI ethical principles on the use of GenAI. Managers need to get clarity from the GenAI providers about fairness, accountability, transparency, accuracy, and autonomy. The developers and designers of

GenAI must understand that GenAI technology should be developed with ethical values so that there is no harm to individuals, organizations or society. The developers need to ensure that the result of the GenAI is non-discriminative, impartial and should not have any bias. Also, the GenAI testers should monitor the accuracy and transparency of GenAI decisions and outputs. Developers can continuously test and validate GenAI outputs against high-quality and well-labelled data.

Managers need to consider these ethical FATAA factors while adopting and implementing GenAI for various business purposes. Managers involved in adopting and implementing GenAI must regularly audit for bias and employ diverse datasets during AI training to ensure fairness. Accountability can be maintained related to GenAI by defining clear roles and responsibilities concerning AI decision-making and outcomes. Top management can develop governance frameworks that assign specific individuals or teams to oversee AI operations and handle any issues that arise. The developers of GenAI and managers can ensure transparency by maintaining clear documentation of AI development processes, decision-making criteria, and data sources. They can communicate openly with stakeholders about GenAI capabilities and limitations. Managers can establish protocols for frequent updates and corrections to maintain the accuracy of GenAI applications. Furthermore, managers can set clear boundaries for GenAI's autonomous operations and involve personnel in overseeing AI systems to ensure its autonomy.

As per this study, innovativeness moderates the association between the use of GenAI and organizational performance (H10) (Acar and Ozs, 2018; Song and Xie, 2000). Hence, managers should be cognizant of organizational innovativeness to achieve organizational performance by using GenAI. In addition, managers can take on a collaborative role in developing GenAI systems to ensure they operate within ethical guidelines, thus preventing unethical issues. Managers can engage themselves as co-design of GenAI and it would also enable the tailoring of GenAI to meet specific organizational requirements. Furthermore, government policymakers can develop policies to regulate GenAI design and implementation by considering the ethical principles discussed, i.e., FATAA. Even IT and ITeS organizations and professional associations can also develop policies for the ethical use of GenAI using the ethical factors addressed in the research.

5.3. Limitations and future research directions

This is the first empirical study that explores the use of GenAI in organizations. However, this study limits itself to the IT and ITeS sectors as a cross-sectional study. Further studies can be conducted in different business sectors, such as banking, education, services and e-commerce. Also, studies can be conducted regarding satisfaction, experience, engagement, and loyalty with GenAI technologies. GenAI is used for various business functions, and investigations can be conducted into its use for marketing, HRM, and supply chain management. Further studies concerning privacy, security, digital divide, and explainability can be undertaken. In this study, since the influence of the use of GenAI on organization performance is solely tied to the number of employees (control variable firm size) in the organization; hence, future studies may consider how effectively GenAI is integrated into particular workflows, quality of available organizational data, capacity of firms to adapt AI applications to firm-specific needs and alignment with organization's goal. Hence, a broader consideration of control variables is required to check the influence on organizational performance. This work is limited to investigating the influence of ethical principles on the use of GenAI. Further studies can be conducted on the impact of these ethical principles on trust on GenAI and, consequently, on the continuing intention to use GenAI. Employees' viewpoints regarding GenAI use can be explored to use it effectively in organizations. Also, how are GenAI developers taking steps to ensure the ethical output and decisions made by GenAI can be investigated in the future.

6. Conclusions

This study utilizes INST and the ethical principles encapsulated by FATAA—(Fairness, Accountability, Transparency, Accuracy, and Autonomy) to examine the use of GenAI and to further analyze its impact on organizational performance. This research is pioneering in that it offers insights into the impact of external pressures and ethical principles on the utilization of GenAI. This study affirms the impact of coercive, normative, and mimetic pressures on the use of GenAI in organizations. It reveals the influence of fairness, accountability, transparency, accuracy and autonomy on the use of GenAI. Further, the moderation effect of organizational innovativeness between the use of GenAI and organizational performance is established in this investigation. This work contributes and enhances the existing body of knowledge by integrating the use of GenAI and examining its ethical principles of GenAI. This research delivers crucial understanding for top-level executives and senior managers in IT and ITeS sectors, directing the ethical application of GenAI to strengthen organizational performance.

CRediT authorship contribution statement

Nripendra P. Rana: Writing – original draft, Conceptualization. **Rajasshrie Pillai:** Writing – review & editing, Writing – original draft, Methodology. **Brijesh Sivathanu:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Nishtha Malik:** Validation, Software, Methodology, Formal analysis.

Data availability

Data will be made available on request.

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